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# EXPERIMENT 7
# Image Captioning using CNN/Transformer Encoder and RNN-style Decoder

import torch
from transformers import VisionEncoderDecoderModel
from transformers import ViTImageProcessor
from transformers import AutoTokenizer
from PIL import Image
from google.colab import files

# Device
device = torch.device(
    "cuda" if torch.cuda.is_available() else "cpu"
)

print("Using device:", device)

# Load pretrained image captioning model
model_name = "nlpconnect/vit-gpt2-image-captioning"

print("Loading model...")

model = VisionEncoderDecoderModel.from_pretrained(model_name)

processor = ViTImageProcessor.from_pretrained(model_name)

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = model.to(device)

model.eval()

print("Model loaded successfully!")

# Upload image
print("\nUpload an image:")

uploaded = files.upload()

image_path = list(uploaded.keys())[0]

print("Selected image:", image_path)

# Open image
image = Image.open(image_path).convert("RGB")

display(image)

# Generate caption
pixel_values = processor(
    images=image,
    return_tensors="pt"
).pixel_values

pixel_values = pixel_values.to(device)

with torch.no_grad():

    output_ids = model.generate(
        pixel_values,
        max_length=30,
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        num_beams=4,  
        early_stopping=True  
    )  
  
# Decode caption  
caption = tokenizer.decode(  
    output_ids[0],  
    skip_special_tokens=True  
)  
  
print("\nGenerated Caption:")  
print(caption)
```

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Using device: cpu
Loading model...
config.json: 100% 4.61k/4.61k [00:00<00:00, 52.8kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable t
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hu
pytorch_model.bin: reconstructing file: 100% 982MB / 982MB, 67.3MB/s
pytorch_model.bin: downloading bytes: 910MB, 54.0MB/s
model.safetensors: reconstructing file: 100% 982MB / 982MB, 65.1MB/s
model.safetensors: downloading bytes: 909MB, 58.2MB/s
Loading weights: 100% 445/445 [00:00<00:00, 4951.49it/s]
[transformers] VisionEncoderDecoderModel LOAD REPORT from: nlpconnect/vit-gpt2-image-captioning
Key | Status | |
-----+-----+--
decoder.transformer.h.{0...11}.crossattention.masked_bias | UNEXPECTED | |
decoder.transformer.h.{0...11}.attn.masked_bias | UNEXPECTED | |

Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect
preprocessor_config.json: 100% 228/228 [00:00<00:00, 2.04kB/s]
tokenizer_config.json: 100% 241/241 [00:00<00:00, 2.01kB/s]
tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 6.76MB/s]
special_tokens_map.json: 100% 120/120 [00:00<00:00, 1.71kB/s]
```